

Three Dimensional Geometry: Formula Sheet & Reference

Class 12 Mathematics – Chapter 11 Complete Reference & Formulas

1 Direction Cosines, Direction Ratios & Lines Joining Two Points

1. Direction Cosines (l, m, n) and Direction Ratios (a, b, c) :

- If a line makes angles α, β, γ with the positive x, y, z -axes:

$$l = \cos \alpha, \quad m = \cos \beta, \quad n = \cos \gamma \implies l^2 + m^2 + n^2 = 1$$

- Relation between Direction Ratios (a, b, c) and Direction Cosines (l, m, n) :

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

2. Direction Cosines of Line Joining Two Points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$:

$$l = \frac{x_2 - x_1}{PQ}, \quad m = \frac{y_2 - y_1}{PQ}, \quad n = \frac{z_2 - z_1}{PQ} \quad \left[PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \right]$$

2 Equations of a Line in Space

1. Line Through a Given Point ($\vec{a}$) and Parallel to $\vec{b}$:

$$\text{Vector Form: } \vec{r} = \vec{a} + \lambda \vec{b}$$

$$\text{Cartesian Form: } \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} \quad (\text{where } \vec{b} = ai + bj + ck)$$

2. Line Passing Through Two Given Points ($P(\vec{a}_1)$ and $Q(\vec{a}_2)$):

$$\text{Vector Form: } \vec{r} = \vec{a}_1 + \lambda(\vec{a}_2 - \vec{a}_1)$$

$$\text{Cartesian Form: } \frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

3 Angle Between Two Lines

1. Vector Form ($l_1 : \vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $l_2 : \vec{r} = \vec{a}_2 + \mu \vec{b}_2$):

$$\cos \theta = \left| \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|} \right|$$

2. Cartesian Form (with Direction Ratios a_1, b_1, c_1 and a_2, b_2, c_2):

$$\cos \theta = \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$$

Conditions for Lines:

- **Perpendicular:** $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$ (or $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$).
- **Parallel:** $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (or $l_1 = l_2, m_1 = m_2, n_1 = n_2$).

4 Shortest Distance Between Lines

1. Shortest Distance Between Two Skew Lines:

- **Vector Form** ($\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$):

$$d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

- **Cartesian Form** ($\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$):

$$d = \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2}}$$

2. Distance Between Two Parallel Lines ($\vec{r} = \vec{a}_1 + \lambda \vec{b}$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}$):

$$d = \left| \frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|} \right|$$